Empirical Foundations of the Evidence-Based Framework for Economic and Social Behavior: A Synthesis of 50 Years of Behavioral Evidence and Modern Measurement Capabilities

Complementarity-Context Framework Collaboration

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The Evidence-Based Framework for Economic and Social Behavior (EBF) integrates half a century of behavioral science research under a unified theoretical structure emphasizing complementarity matrices C and context-dependent parameterizations Ψ. This paper synthesizes the empirical foundations that support the framework, demonstrating that core concepts are not merely logically coherent but empirically real, measurable, and consequential for policy intervention.

The analysis proceeds through five research programs: (1) experimental economics of social preferences, establishing that complementarities in utility functions are widespread and robust across methodological variants; (2) field evidence demonstrating external validity of laboratory findings across diverse settings; (3) replication studies confirming core findings despite systematic effect-size modulation by context; (4) complexity economics and network effects documenting complementarity structures at aggregate scale; and (5) neuroeconomic and cross-cultural evidence establishing biological and cultural universality of the framework structure with systematic parametric variation.

A diagnostic protocol is introduced that transforms empirical deviations from framework predictions into actionable research signals, enabling self-correction and progressive refinement. Technological capabilities emerging 2020-2026—high-dimensional data, computational structural estimation, natural language processing, and large language models—have rendered tractable what was previously infeasible: simultaneous estimation of context-dependent complementarity matrices across multiple domains.

The paper concludes that the framework represents a progressive research program in the Lakatosian sense: a protective core (the 8C structure) with an adaptive auxiliary belt (context parameterizations and measurement methods) that can be systematically refined in response to empirical evidence without abandoning fundamental principles.

Keywords: complementarity, context, behavioral economics, experimental evidence, structural estimation, framework integration, falsifiable predictions

Contents

Chapter 1

Introduction: Why Empirical Foundations Matter

The theoretical apparatus developed in contemporary behavioral economics—complementarity matrices, context dynamics, coherence indices—risks remaining abstract unless grounded in empirical evidence. This paper establishes that the core concepts of the Evidence-Based Framework are not merely logically possible but empirically real, measurable, and consequential for predicting and shaping economic behavior.

The challenge confronting theoretical behavioral economics is distinct from that facing traditional neoclassical theory. The latter could claim that preferences are revealed through choice; behavioral economics must demonstrate that non-standard preference structures actually characterize observed behavior. This is both more ambitious and more vulnerable to empirical falsification.

Before reviewing the evidence systematically, we address a foundational methodological question: *How should empirical evidence be integrated into predictive frameworks?* Recent work in machine learning and large language models has revealed a critical limitation: knowing *about* behavioral phenomena—understanding their theoretical structure—is insufficient for *predicting* them. The distinction between knowing and predicting drives the analysis that follows.

The argument proceeds as follows. Section **??** establishes why *calibration on experimental data*, rather than training on text descriptions, is necessary for quantitative behavioral prediction. Sections **??** through **??** then marshal evidence from five independent research traditions—all pointing toward the same conclusion: complementarities are real, context-dependent, and operationalizable. Section **??** introduces a diagnostic protocol that transforms empirical deviations from predictions into signals for framework refinement. Section **??** documents the technological capabilities that have made the empirical program outlined

herein tractable. The paper concludes with implications for policy intervention and future research.

Chapter 2

Calibration, Not Simulation: Methodological Foundations

2.1 The Prediction Problem

Contemporary large language models—trained on text corpora encoding decades of behavioral economics literature—can articulate the logic of complementarity. When prompted about the Ultimatum Game, an LLM can state: "Players exhibit inequality aversion, rejecting unequal offers even at personal cost. This suggests utility depends not only on own payoff but on the distribution of payoffs."

Yet when asked for *quantitative predictions*—"What fraction of Responders will reject an offer of 20% of the pie?"—the same model's predictions are often inaccurate. Why? Knowing the *structure* of behavior is not equivalent to knowing its *magnitude*.

Formal Statement of the Problem. Let Y denote observed behavior (e.g., rejection rates), $\mathcal{T}$ denote a theoretical model (e.g., Fehr-Schmidt inequity aversion), and $\boldsymbol{\theta}$ denote parameters of that model (e.g., α, β capturing inequality aversion). The prediction problem has two stages:

$$\text{Stage 1:} \quad Y = \mathcal{T}(\boldsymbol{\theta}, \mathcal{X}) + \epsilon \tag{2.1}$$

where $\mathcal{X}$ are contextual variables and ϵ represents prediction error.

$$\text{Stage 2:} \quad \hat{\mathcal{T}}(\boldsymbol{\theta}^*, \mathcal{X}') \approx Y' \tag{2.2}$$

where $\boldsymbol{\theta}^*$ are estimated parameters and Y' are new observations.

LLMs trained on behavioral economics literature can often correctly specify Equation **??**.

They can articulate the structural relationship between behavior and underlying preferences. However, they systematically fail at Stage 2 because they lack calibrated parameter estimates $\boldsymbol{\theta}^*$.

Core Concept: Calibration vs. Simulation

Simulation means: Given a theoretical structure $\mathcal{T}$ a
erate predictions $\hat{Y} = \mathcal{T}(\boldsymbol{\theta}, \mathcal{X})$.
Calibration means: Given empirical data $\{(Y_i, \mathcal{X}_i)\}_{i=}^{n}$
prediction error:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} (Y_i - \mathcal{T}(\boldsymbol{\theta}, \mathcal{X}$$

The distinction matters because:

- **Simulation with wrong parameters**: Structu
inaccurate
- **Calibration on experimental data**: Structur
accurate

The Calibration-Simulation Distinction.

2.2 Parameter Estimation: The Three-Tier Hierarchy

The framework employs a three-tier hierarchy for parameter estimation, detailed in Appendix BBB. The tiers are distinguished by information quality and cost.

2.2.1 Tier 1: Empirical - Randomized Controlled Experiments

The goldstandard for parameter estimation is data from experiments designed specifically to identify parameters of theoretical interest. For inequity aversion, the Fehr-Schmidt parameters α (disadvantageous inequity aversion) and β (advantageous inequity aversion) have been estimated across dozens of studies.

Tier 1 estimation proceeds as follows:

1. **Experiment Design**: Researchers conduct experiments with payoff structures chosen to identify specific parameters. For Fehr-Schmidt, ultimatum games, dictator games, and allocation choice experiments are standard.

2. **Data Collection**: Typically $n = 50$–500 subjects (individuals or lab sessions) across multiple parameter configurations.

3. **Structural Estimation**: Maximum likelihood estimation or Bayesian methods recover $\hat{\boldsymbol{\theta}}$. Standard errors reflect sampling uncertainty.

4. **Replication**: The same study is conducted in multiple labs, countries, subject pools, and time periods. Consistency of estimates across replications indicates robustness.

2.2.2 Tier 2: Meta-Analytical - Synthesis Across Heterogeneous Studies

Not all parameters can be directly estimated through controlled experiments. For instance, the complementarity between institutional legitimacy and tax compliance is difficult to manipulate experimentally. For such cases, Tier 2 uses meta-analytical synthesis.

Tier 2 analysis follows a structured protocol:

1. **Literature Identification**: Systematic search identifies all studies containing information relevant to parameter estimation. Study selection criteria are pre-specified.

2. **Effect Extraction**: From each study, key statistics (point estimates, standard errors, sample sizes) are extracted. Heterogeneity in operationalization is documented.

3. **Meta-Regression**: A regression model relates study characteristics (year, country, methodology) to effect sizes. This enables prediction of parameters under conditions not directly observed.

4. **Quality Assessment**: Bias risk is assessed using standard instruments (Cochrane Risk of Bias, ROBINS-I). Estimates are sensitivity-tested against assumptions.

Appendix EV exemplifies this approach, conducting a meta-analysis of 500+ experiments on complementarity across 8 paradigms (Social Preferences, Honesty, Market Experiments, Market Design, Coordination, Learning, Repeated Games, Beliefs).

2.2.3 Tier 3: Synthetic Priors - Large Language Models as Auxiliary Estimators

Some parameters cannot be estimated through Tiers 1 or 2. This occurs when:

- Relevant studies exist but are qualitative or report insufficient statistics

- Context-specific parameters (e.g., complementarity in a novel domain) lack direct evidence

- Temporal urgency prevents waiting for empirical research

For these cases, Tier 3 uses Large Language Models with structured elicitation, documented in Appendix AN (LLM Monte Carlo methodology).

The key is that Tier 3 is an *auxiliary* method that complements—rather than replaces—Tiers 1 and 2. LLM estimates are validated against empirical benchmarks. Appendix AN reports 6 internal validation tests, all indicating that LLM-derived parameters correlate robustly with experimental estimates (mean $r = 0.87$, $p < 0.0001$).

2.3 Why This Matters for the Framework

The three-tier hierarchy addresses a fundamental tension in behavioral economics:

> *"We want to build quantitative models that generate point predictions, yet experimental estimation of all parameters is infeasible. How do we proceed without introducing bias?"*

The answer is structured hierarchy. Tier 1 provides gold-standard estimates. Tier 2 extends those estimates to related contexts through meta-analysis. Tier 3 provides auxiliary estimates for parameters where direct evidence is unavailable, validated against Tier 1 benchmarks.

This approach enables the framework to remain falsifiable: predictions can be tested against data, and parameter values can be revised when predictions fail. By contrast, an LLM-only approach (describing phenomena without calibration) sacrifices falsifiability.

Chapter 3

The Experimental Revolution: Social Preferences and Complementarity

The experimental study of social preferences—how people value not only their own payoff but also the welfare of others—provides the richest evidence for complementarity in utility functions. Four experimental paradigms stand out: the Ultimatum Game, Dictator Game, Public Goods Games, and Trust Games. Each reveals a distinct aspect of how individual utilities interact.

The evidence reviewed below comes from Appendix AJ (Social Preferences: Other-Regarding Behavior), which synthesizes 20+ foundational papers with 70,000+ combined citations, and Appendix EV (Systematic Review of Experimental Evidence), which conducts meta-analysis of 500+ experiments across 8 paradigms.

3.1 The Ultimatum Game: Inequality Aversion Under Stakes

3.1.1 Experiment and Prediction

The Ultimatum Game, introduced by **?**, provides perhaps the cleanest test of whether utility depends on the distribution of payoffs.

Protocol. Two players divide a monetary sum (e.g., \$10):

- **Proposer** chooses a split (x_p, x_r) where $x_p + x_r = 10$.
- **Responder** observes the proposal and either accepts (both receive as proposed) or rejects (both receive \$0).

Standard Economic Prediction. If preferences are separable—i.e., $U_i(x_i, x_{-i}) = u_i(x_i)$ where u_i is a function of own payoff only—then:

1. Responders should accept any positive offer (since \$0.01 ¿ \$0).
2. Proposers, anticipating this, should offer the minimum divisible unit.
3. Subgame perfect equilibrium: Proposer offers \$0.01 to Responder.

This prediction follows from purely selfish preferences, without additional assumptions about irrationality or uncertainty.

3.1.2 Actual Results

? and hundreds of subsequent replications (meta-analyzed in Appendix EV) document robust deviations from this prediction. Across diverse contexts:

Table 3.1: Ultimatum Game Results: Modal Behavior Across 50+ Replications

	Parameter	**Observed Value**
Proposer Behavior	Modal offer	40–50% of pie
	Mean offer	30–40% of pie
	Std. Dev.	10–20 percentage points
Responder Behavior	% rejecting 20% offer	50%+
	% rejecting 10% offer	80%+
	% rejecting 1% offer	95%+

What This Reveals. The rejection of low offers is costly to Responders (they forego positive payoff). Yet many Responders incur this cost rather than accept what they perceive as unfair allocations. This behavior is inconsistent with separable preferences but consistent with complementarity in the utility function:

$$\frac{\partial^2 U_i}{\partial x_i \partial x_j} \neq 0 \tag{3.1}$$

That is, Responder's utility depends jointly on own payoff (x_i) and the distribution (x_j).

3.1.3 Formalizing Inequality Aversion

? proposed a specific functional form for complementarity in preferences:

$$U_i(\mathbf{x}) = x_i - \alpha_i \cdot \frac{1}{n-1} \sum_{j \neq i} \max(x_j - x_i, 0) - \beta_i \cdot \frac{1}{n-1} \sum_{j \neq i} \max(x_i - x_j, 0) \tag{3.2}$$

The two complementarity parameters are:

- α_i: **Disadvantageous Inequity Aversion** — disutility from earning less than others (envy)
- β_i: **Advantageous Inequity Aversion** — disutility from earning more than others (guilt)

Estimated Parameters (Tier 1 Evidence). From experimental data directly estimating Equation **??**:

Table 3.2: Fehr-Schmidt Parameter Estimates from Experimental Data

	Mean	**Median**	**Min**	**Max**
α (envy)	0.85	0.82	0.00	4.20
β (guilt)	0.31	0.28	0.00	0.60
Constraint: $\beta \leq \alpha$	Observed in 94% of estimates			

These estimates reveal two robust facts:

1. **High Heterogeneity:** Both α and β vary dramatically across individuals (CV ≈ 1.0). Some people are nearly selfish ($\alpha, \beta \approx 0$), while others incur large costs to punish unfairness.
2. **Envy ¿ Guilt:** The constraint $\beta \leq \alpha$ is nearly universal. People dislike disadvantageous inequality more intensely than advantageous inequality—consistent with asymmetric loss aversion in prospect theory.

3.2 The Dictator Game: Pure Social Preferences

3.2.1 Experiment and Prediction

The Dictator Game removes the Responder's veto power. The Proposer simply chooses the allocation, and that choice is implemented without possibility of rejection.

Standard Prediction. Without strategic reasons to share, a selfish Proposer should take the entire amount, leaving $0 for the Dictator.

3.2.2 Actual Results

? and subsequent replications find:

Table 3.3: Dictator Game Results: Mean Giving Across 40+ Replications

	Statistic
Mean transfer (fraction of pie)	20–30%
Fraction giving 0%	30–40%
Fraction giving 50%	10–20%
Fraction giving 100%	1–3%

This is remarkable: even without reciprocal incentives or punishment risk, substantial majorities of Proposers voluntarily transfer money to anonymous others. This demonstrates pure social preferences: utility explicitly depends on others' welfare.

Interpretation. The Dictator Game rules out several explanations for Ultimatum Game behavior:

- **Not strategic:** No threat of rejection, yet sharing persists
- **Not norm-learning:** Anonymous decision, no repeated interaction
- **Not punishment fear:** Third party cannot punish, Dictator has no future reputation concerns

This leaves only one conclusion: utility functions genuinely incorporate others' welfare.

3.3 Public Goods Games: Strategic Complementarity

3.3.1 Experiment and Prediction

The Public Goods Game, pioneered by **?**, tests complementarity at the group level.

Protocol. n players each receive endowment e. Each simultaneously chooses contribution $c_i \in [0, e]$ to a common pool. The pool is multiplied by $m > n$ and distributed equally:

$$\pi_i = e - c_i + \frac{m \sum_j c_j}{n} \tag{3.3}$$

The private return to contributing is $\frac{m}{n} - 1 < 0$. Nash equilibrium: $c_i^* = 0$ for all i.

3.3.2 Actual Results

Table **??** summarizes findings from meta-analysis of 100+ public goods experiments:

Table 3.4: Public Goods Games: Contribution Behavior Over Time

	Round 1–3	**Round 10–15**
Mean contribution (fraction of e)	0.50	0.15
Std. Dev.	0.20	0.15
% Conditional Cooperators	50%	45%
% Free Riders	30%	40%
% Unconditional Cooperators	10%	5%

3.3.3 A Puzzle: Conditional Cooperation

The most striking finding is that contributions decline over repeated rounds but stabilize around 15–20% rather than falling to zero. Survey data reveals the reason: most contributors are **conditional cooperators**. They contribute because they expect others to contribute.

Formally, conditional cooperation is strategic complementarity:

$$\frac{\partial c_i}{\partial \bar{c}_{-i}} > 0 \tag{3.4}$$

where $\bar{c}_{-i}$ is the mean contribution of others. An individual's contribution increases with others' contributions—a complementary relationship.

Evidence for Complementarity Structure. This complementarity is not fixed but context-dependent. When the game includes a punishment option (second-mover stage where players can punish free-riders), contributions rise from 15% to 80%+ and remain high across repeated rounds. This demonstrates $C_{cooperation}(\Psi) = C_{cooperation}(\Psi_{punishment})$: the complementarity structure depends on the institutional context.

3.4 Trust Games: Reciprocity Quantified

3.4.1 Experiment and Prediction

The Trust Game, introduced by **?**, quantifies reciprocity as a complementary relationship between Sender's trust and Receiver's trustworthiness.

Protocol.

1. Sender receives initial endowment (e.g., \$10)
2. Sender can send $s \in [0, 10]$ to Receiver
3. Sent amount is tripled to $3s$
4. Receiver can return $r \in [0, 3s]$ to Sender
5. Final payoffs: Sender gets $10 - s + r$; Receiver gets $3s - r$

3.4.2 Standard Prediction

Under selfish preferences:

1. Receiver has no incentive to return anything: $r^* = 0$
2. Sender, anticipating this, sends nothing: $s^* = 0$
3. Equilibrium: Complete absence of trust and exchange

3.4.3 Actual Results

? and 100+ subsequent studies find:

Table 3.5: Trust Game Results: Reciprocal Behavior

	Median/Mean
Sender: Amount sent (fraction of e)	0.50
Receiver: Amount returned (fraction of received)	0.92
Correlation: $r_{(s,r)}$	0.68–0.72

The correlation between sent and returned amounts demonstrates reciprocity as a quantifiable complementarity: more trust triggers more trustworthiness.

Formal Structure. The complementarity can be written as:

$$r = f(s) \quad \text{where} \quad f'(s) > 0, \quad f''(s) \lesseqgtr 0 \tag{3.5}$$

The positive first derivative $f'(s) > 0$ indicates complementarity: Receiver's return is increasing in Sender's trust.

3.5 Meta-Analytical Synthesis: Across 500+ Experiments

3.5.1 The Scope of Appendix EV

Appendix EV conducts systematic meta-analysis of complementarity across 8 experimental paradigms:

1. Social Preferences (Ultimatum, Dictator, Allocation Choice)
2. Honesty (Die Roll, Coin Flip, Self-Reporting Tasks)
3. Market Experiments (Posted Offer, Double Auction, Bilateral Bargaining)
4. Market Design (Voting, Auctions, Matching Markets)
5. Coordination (Minimum Effort Games, Stag Hunt, Assurance Games)
6. Learning (Guessing Games, Beauty Contest, Information Cascades)
7. Repeated Games (Prisoner's Dilemma, Matching Pennies)
8. Beliefs (Forecasting, Surveys, Stated Preferences)

Across these 8 paradigms, the analysis identifies:

- 500+ experiments with sample sizes totaling 50,000+ individual decisions
- Effect sizes for each paradigm (mean difference from null hypothesis)
- Heterogeneity sources (subject pool, stakes, country, year)
- Correlation structure between effect sizes across paradigms

3.5.2 Core Findings

Meta-regression analysis yields three central conclusions:

Finding 1: Complementarities are Ubiquitous. Effect sizes are significantly different from zero in all 8 paradigms. That is,

$$P\left(\frac{|\bar{C}|}{SE(\bar{C})} > 1.96\right) > 0.95 \quad \text{for all 8 paradigms} \tag{3.6}$$

The evidence is overwhelming that complementarity—the condition $\frac{\partial^2 U_i}{\partial x_i \partial x_j} \neq 0$—is the rule rather than the exception in human preferences.

Finding 2: Effect Sizes are Heterogeneous and Context-Dependent. Effect sizes vary systematically with:

- **Stakes**: Doubling monetary stakes reduces effect size by 5–15%
- **Subject Pool**: Students average effect size 0.8 SD; non-students average 0.6 SD
- **Country**: Effect sizes correlate with institutional quality and social trust
- **Paradigm**: Effect sizes range 0.3–0.9 SD across paradigms

This heterogeneity is not noise but structure. It supports the framework's claim that complementarity is context-dependent: $C_{ij} = C_{ij}(\Psi)$.

Finding 3: Effects Replicate Reliably. The replication crisis in psychology—where many classic findings could not be reproduced—did not undermine behavioral economics. Camerer et al. (2016) replicated 21 landmark experimental economics studies with preregistration and resource constraints typical of replications.

Results:

- 62% replicated with same direction and significance
- 77% replicated with same direction (allowing smaller effect sizes)
- Average effect size: 75% of original

The core findings—inequality aversion, conditional cooperation, reciprocity, trust—survive methodological scrutiny. This robustness strengthens the claim that these are genuine features of human preferences, not experimental artifacts.

Chapter 4

Field Evidence and External Validity

4.1 The Field-Lab Gap: Does it Exist?

A persistent critique of experimental economics questions whether laboratory findings generalize to real-world settings where stakes are higher, reputation concerns are salient, and institutional contexts are complex.

4.1.1 The Critique

Levitt and List (2007) articulated concerns on three fronts:

1. **Stakes effects**: Do people behave differently when money is "real" and large?
2. **Subject pool**: Are students representative of broader populations?
3. **Artificiality**: Do laboratory protocols (anonymity, abstract framing) remove features essential to real-world behavior?

4.1.2 The Evidence

Subsequent research has investigated each concern:

Stakes Effects. **?** and meta-analyses suggest that stakes effects are real but moderate:

- Doubling stakes increases selfish behavior slightly (effect: 5–15% reduction in cooperation)
- Effect size is small relative to the cooperation-to-selfishness gap documented in experiments

- Pattern holds across paradigms (Dictator, Public Goods, Trust Games)

The interpretation is that stakes matter but do not overturn core findings. Social preferences persist even with high stakes, though magnitude may modulate.

Subject Pool. Studies comparing students to non-students find:

- Non-students are slightly more prosocial on average (2–8% higher cooperation)
- Variance across both groups is substantial (overlapping distributions)
- Relative ordering of individuals is stable across demographic groups

Again, the gap exists but is modest compared to the effect sizes of social preferences themselves.

Lab vs. Field Experiments. A growing body of field experiments tests laboratory predictions in real-world contexts:

- **Gift Exchange in Labor: ?** and **?** conducted field experiments on workers. Workers who received unexpectedly high wages increased effort by 10–20%. This mirrors gift-exchange reciprocity documented in labs.
- **Charitable Giving:** Field experiments on donation matching (**?**) show that social preferences (desire to reciprocate generosity) affect giving behavior in real fundraising campaigns. Effects are substantial (doubling effect sizes similar to lab).
- **Tax Compliance: ?** and **?** provide evidence that tax compliance depends on perceived institutional fairness and beliefs about others' compliance. This is conditional cooperation at scale.

4.2 Trust and Economic Development

4.2.1 The Correlation

A robust cross-country correlation links generalized trust to GDP per capita:

$$\text{GDP per capita}_c = \beta_0 + \beta_1 \cdot \text{Trust}_c + \epsilon_c \tag{4.1}$$

Meta-analysis of cross-national datasets (World Values Survey, OECD, etc.) yields $\beta_1 \approx 0.6$ (correlation between trust and log GDP). This suggests that countries with higher social trust are wealthier.

4.2.2 Causality Question

The natural question is whether trust *causes* economic development or vice versa. Three approaches address this:

Approach 1: Inherited Trust among Immigrants. **?** exploit an elegant natural experiment. U.S. immigrants from different countries inherit different levels of generalized trust from their countries of origin. If trust causes economic behavior, immigrants from high-trust countries should exhibit more cooperative behavior in the U.S. than immigrants from low-trust countries, even though all face the same institutional environment.

Results:

- Inherited trust predicts participation in groups, civic engagement, and cooperation
- Effect persists for first-generation immigrants and attenuates (but remains significant) for second generation
- Effect size suggests trust explains 15–20% of variation in cooperative behavior

Approach 2: Institutional Reform and Trust. **?** examine how institutional reforms change trust. When countries improve institutional quality (rule of law, democratic accountability), trust increases in subsequent surveys.

Approach 3: Experimental Evidence from High- and Low-Trust Societies. **?** conduct trust game experiments across 9 countries with varying institutional quality. Key finding:

$$\text{Trust}_{game,c} = \alpha + \beta \cdot \text{Institutional Quality}_c + \gamma \cdot \text{Country Baseline Trust}_c + \epsilon_c \tag{4.2}$$

Both institutional quality and baseline trust predict game-based trust behavior. The correlation structure suggests complementarity: institutional quality and behavioral trust reinforce each other.

4.3 Cooperation and Common Pool Resources

? documented a striking phenomenon: communities manage common pool resources (fisheries, forests, irrigation systems) without formal government regulation or privatization. They develop self-enforcing norms and institutions that sustain cooperation.

This provides field evidence for strategic complementarity at scale. Successful commons regimes feature:

1. **Monitoring institutions**: Clear rules and monitoring of compliance
2. **Graduated sanctions**: Punishment increasing with severity of violation
3. **Conflict resolution**: Mechanisms for dispute resolution
4. **Community homogeneity**: Easier trust and norm enforcement

Each of these is a complementarity:

- Monitoring makes punishment more effective (complementarity between monitoring and enforcement)
- Punishment makes rule-following more attractive (complementarity between rules and compliance)
- Community homogeneity makes trust easier (complementarity between similarity and trust)

The success of commons management provides evidence that complementarity structures can be *endogenously constructed* through institutional choice.

Chapter 5

Complexity Economics and Aggregate Complementarities

5.1 Increasing Returns and Path Dependence

? demonstrated that complementarity at the aggregate level (positive feedback between adoption decisions) generates phenomena impossible under standard diminishing returns models:

- **Path Dependence**: History matters; small initial advantages compound
- **Lock-In**: Inefficient technologies can dominate once complementarity structure is established
- **Multiple Equilibria**: Identical fundamentals can support different outcomes

Example: The QWERTY keyboard layout. This layout is suboptimal for typing speed (Dvorak layout is faster), yet QWERTY dominates globally. Why? Complementarities:

- Workers have already learned QWERTY (high switching cost)
- Keyboard manufacturers produce QWERTY machines (network effect)
- Software and documents assume QWERTY layout (compatibility)

These complementarities create lock-in: switching to Dvorak requires coordinated change across millions of people. The equilibrium is *not* a stable rest point but a self-reinforcing configuration.

5.2 Network Effects and Winner-Take-All

? formalized network effects. If utility from adopting a technology increases with the number of other adopters:

$$U_i(adopt) = v(N) - p \quad \text{where} \quad N = \sum_j a_j, \quad v'(N) > 0 \tag{5.1}$$

then adoption decisions are strategic complements:

$$\frac{\partial}{\partial a_{-i}} (\mathbb{1}[U_i(adopt) > U_i(not)]) > 0 \tag{5.2}$$

This generates:

- **Tipping Points**: Small initial differences become large
- **Winner-Take-All**: One network dominates (Facebook vs. Myspace)
- **Hysteresis**: Hard to reverse once established

The empirical evidence is overwhelming. Digital markets (mobile operating systems, social networks, cloud platforms) display classic winner-take-all dynamics driven by complementarities.

5.3 Santa Fe Perspective: Economy as Adaptive System

? and the Santa Fe Institute have reframed economics as the study of **adaptive systems**:

> *"The economy is not a system in equilibrium. It is a system that is perpetually constructing itself. Agents form expectations and act on them, which affects what happens, which changes agents' expectations. The economy is coherent in behavior but adaptive in its structure."*

This perspective aligns with the framework's notion of stability as coherence rather than equilibrium:

$$K = \frac{\|C^*(\Psi) - C(\Psi)\|}{\max \|C^*(\Psi)\|} \approx 1 \quad \text{(coherence)} \tag{5.3}$$

The economy can be stable (coherent) while agents' expectations and behaviors are continuously adjusting. This explains why complementarity structures are robust ($K \approx 1$) yet adaptive.

Chapter 6

Neuroeconomics and Biological Substrates

6.1 Social Preferences and Neural Activation

? and **?** demonstrated using fMRI that social rewards and material rewards activate overlapping neural circuits.

Key Finding. When subjects play cooperation games with variable levels of fairness:

- Fair outcomes (equal split) activate reward circuitry (ventral striatum, OFC)
- Unfair outcomes activate aversion circuitry (anterior insula, dorsolateral prefrontal cortex)
- Social and material rewards are encoded in a common currency in ventromedial prefrontal cortex (vmPFC)

Implication for FEPSDE. The framework posits that utility has 6 dimensions: Financial (F), Emotional (E), Physical (P), Social (S), Digital (D), Ecological (E-cology). Neuroscientific evidence shows these are processed by overlapping but distinct neural systems:

- **Common Currency**: All dimensions converge on vmPFC, suggesting they are combined in a unitary utility function
- **Distinct Substrates**: Each dimension has dedicated processing regions before convergence (e.g., social reward $\rightarrow$ superior temporal sulcus; pain $\rightarrow$ anterior insula)

- **Substitutability**: The same level of utility can be achieved by different combinations of dimensions (e.g., money or status can substitute for social approval)

This biological evidence supports the framework's dimensional structure.

6.2 Oxytocin, Neuromodulation, and Context

? demonstrated that neuroendocrine states modulate social preferences. Subjects given intranasal oxytocin vs. placebo played trust games:

Table 6.1: Oxytocin Effects on Trust Game Behavior

	Oxytocin	**Placebo**
Mean amount sent (fraction of e)	0.63	0.54
Effect size (Cohen's d)	0.87 (large)	
Non-social task (risk game)	No effect	

Interpretation: Oxytocin specifically increases trust in social contexts, not general risk-taking. This demonstrates that complementarity structures have biological modulators:

$$C_{trust}(\Psi_{oxytocin}) > C_{trust}(\Psi_{placebo}) \tag{6.1}$$

6.3 Third-Party Punishment and Intrinsic Motivation

? found that punishing free-riders activates the reward system (caudate nucleus, striatum) even when punishment is costly. People derive satisfaction from enforcing norms.

This provides a biological mechanism for why complementarity structures can be self-sustaining:

- Norm-following is rewarding (reinforces continued compliance)
- Punishment of defectors is rewarding (reinforces enforcement)
- High cooperators and punishers gain psychological reward even without material benefit

This explains field evidence showing that cooperation can be maintained at large scale through peer punishment, despite free-rider incentives.

Chapter 7

Cross-Cultural Evidence and Universality

7.1 The WEIRD Problem

? documented that the vast majority of behavioral science research recruits subjects from Western, Educated, Industrialized, Rich, Democratic (WEIRD) societies. These populations are often statistical outliers on measures of individualism, time preference, and risk aversion.

The question is whether complementarity structures—the fundamental mechanism proposed by the framework—are universal or culturally specific.

7.2 Cross-Cultural Ultimatum Games

? conducted ultimatum games in 15 small-scale societies across diverse economic and institutional contexts:

Table 7.1: Ultimatum Game Offers Across 15 Societies

Society	**Mean Offer (% of pie)**	**Rejection Rate**
Machiguenga (Peru)	26%	0%
Tsimane (Bolivia)	32%	5%
Hadza (Tanzania)	34%	10%
Kapauku (Indonesia)	45%	4%
Lamelara (Indonesia)	58%	25%
Mean across 15 societies	37%	9%

Key Observations.

1. **Offers above zero**: All societies show positive offers, inconsistent with pure selfishness. This is universal evidence for social preferences.

2. **Substantial variation**: Offers range from 26% to 58%, suggesting context-dependence of complementarity magnitude.

3. **Correlation with market integration**: Offers correlate with market integration ($r = 0.45$) and cultural institutions supporting fairness. The *structure* of social preferences is universal; the *parameters* vary with context.

Theoretical Implication. The data support the framework's claim of universal structure with context-dependent parameters:

$$C_{ij}^{\text{universal}} \text{ but } C_{ij} = C_{ij}(\Psi_{market,culture,institutions}) \tag{7.1}$$

7.3 Culture and Antisocial Punishment

? discovered a phenomenon absent in WEIRD samples: "antisocial punishment" where high contributors are sanctioned alongside free-riders.

In low-trust societies, public cooperation can be perceived as threatening (showing off, making others look bad). This generates negative complementarity:

$$C_{ij} < 0 \quad \text{(strategic substitutes, not complements)} \tag{7.2}$$

This demonstrates that complementarity structure itself—whether $C_{ij} > 0$ or $C_{ij} < 0$—can vary with institutional and cultural context.

Chapter 8

A Diagnostic Protocol: Learning from Deviations

The sections above establish that complementarities are real, measured, context-dependent, and nearly universal. But what happens when observed behavior C deviates from the reference structure $C^*(\Psi)$?

Rather than treating such deviations as noise, the framework employs them as diagnostic signals that reveal incompleteness, measurement failure, or systems in transition.

8.1 Classification of Deviations

Deviations are classified into five types, each with distinct implications:

Type I: Random Deviations. If residuals $\hat{\epsilon}_i = Y_i - \mathcal{T}(\boldsymbol{\theta}^*, \Psi_i)$ are uncorrelated with any measured variables, they represent measurement error. The response is methodological: improve data quality, increase sample size, or refine measurement instruments.

Type II: Context-Systematic Deviations. If residuals correlate with one or more context variables Ψ_k that were not included in $C^*(\Psi)$, this signals a missing interaction term. For instance:

> "Cooperation is lower than predicted in low-trust societies $({}_S low) and high economic inequality ({}_I high) s$
> Ψ_I may amplify complementarity reduction."

Response: Add interaction terms to $C^*(\Psi)$ and re-estimate.

Table 8.1: Taxonomy of Deviations from Framework Predictions

Type	**Signature**	**Interpretation**
I: Random	Uncorrelated with Ψ	Measurement noise; improve instruments
II: Ψ-Systematic	Correlates with Ψ_k	Missing interaction; enrich $C^*(\Psi)$
III: Extra-Ψ	Correlates with $Z \notin \Psi$	Candidate 9th dimension; test primitiveness
IV: Temporal	Autocorrelated	System in transition; model dynamics
V: Localized	Concentrated in specific units	Idiosyncratic factors; conduct case study

Type III: Extra-Ψ Deviations. If residuals correlate with a variable Z that is neither C, Ψ, nor known to influence them, this suggests a candidate 9th context dimension. The framework currently posits 8 dimensions: $\Psi = (\Psi_S, \Psi_I, \Psi_T, \Psi_C, \Psi_E, \Psi_N, \Psi_A, \Psi_R)$.

If, say, ethnic fractionalization consistently predicts residuals not explained by the 8 dimensions, this could indicate a new primitive dimension. The response is a primitiveness test:

1. Does Z operate independently or through existing Ψ_k?
2. Is Z fundamental or derived from other factors?
3. Does adding Z improve predictive power beyond the 8 dimensions?

Type IV: Temporal Deviations. If residuals are autocorrelated—large positive errors followed by large negative errors—the system may be in transition. This is common in post-crisis periods where complementarity structures are rewiring.

Response: Model the dynamics explicitly. How quickly does $C(t)$ adapt toward $C^*(\Psi(t))$?

Type V: Localized Deviations. If deviations are concentrated in specific countries, sectors, or groups, local factors may dominate. Response: Detailed case study of the anomalous unit, potentially revealing context variables not yet included in Ψ.

8.2 Examples from Applied Research

Example 1: East Asia Education Returns (Type II). Standard models predict that education returns to human capital decline with more extensive education (concavity). Yet East Asian countries exhibit higher-than-predicted returns to tertiary education. A Type II diagnostic reveals that this is not random but systematic: correlates with high cultural value for education (Ψ_C high) *and* strong peer effects in schools (Ψ_S high).

The interaction $\Psi_C \times \Psi_S$ amplifies complementarity, producing above-predicted returns. Response: Model interaction term.

Example 2: Institutional Hysteresis (Type IV). Post-financial crisis (2008-2010), institutional trust declined sharply. Standard models predict rapid recovery as institutions improve. Yet trust recovered slowly, with significant autocorrelation in deviations. This Type IV signature suggests hysteresis: past damage to trust affects current expectations about future trust quality.

Response: Model trust dynamics with lagged terms and adjust predictions for institutional "scarring" effect.

Example 3: Ethnic Fractionalization (Type III). Ethnic fractionalization predicts public goods cooperation levels beyond what the 8 context dimensions explain. Response: Test whether ethnic fractionalization operates *directly* as a primitive dimension, or indirectly through reduced social trust (Ψ_S) and institutional quality (Ψ_I).

Results support the indirect pathway: ethnic fractionalization $\rightarrow \Psi_S$ and $\Psi_I \rightarrow$ cooperation. Thus ethnic fractionalization is not a primitive 9th dimension but operates through existing factors.

Chapter 9

Technological Enablement: Why Measurement Is Tractable Now

The empirical foundations documented above have existed for decades. What has changed—radically—is our ability to *measure at scale* and *integrate evidence* across domains.

9.1 The Traditional Limitations

Before 2010, the empirical program outlined herein faced severe constraints:

- **Laboratory experiments**: Small samples ($n = 30$–300), limited contexts
- **Surveys**: Self-report bias, costly ($50–$200 per respondent), slow
- **Field studies**: Case-by-case, difficult to aggregate to cross-national patterns
- **Cross-cultural research**: Expensive, slow, sparse coverage (5–10 countries)

Estimating a 144×144 complementarity matrix C under these constraints was infeasible.

9.2 Digital Behavioral Data (2010–Present)

Every online interaction generates data about revealed preferences:

- **E-commerce**: Purchasing patterns reveal financial and preference complementarities (C^{FF})
- **Social Media**: Social interaction patterns reveal social complementarities (C^{SS})

- **Health Apps**: Activity, sleep, nutrition data reveal physical utility patterns (C^{PP})
- **Location Data**: Movement patterns reveal social, recreational, and ecological complementarities (C^{SS}, C^{DD}, C^{EE})
- **Financial Data**: Credit, savings, investment behavior reveal financial and temporal complementarities (C^{FF}, C^{FT})

The scale is unprecedented: billions of observations across millions of individuals in diverse contexts, enabling estimation of context-dependent complementarities.

9.3 Natural Language Processing for Soft Variables

Context Ψ includes institutional factors that resist direct measurement: norms, trust, legitimacy, fairness. NLP and text-as-data methods enable:

- **Sentiment Analysis**: Institutional legitimacy measured from policy discussion and media coverage (**?**)
- **Topic Modeling**: Identify dominant policy discourse and framing effects
- **Embedding-Based Similarity**: Measure cultural distance between societies using language corpora
- **Automated Legal Coding**: Extract institutional features from regulatory texts

9.4 Large Language Models: Structural Elicitation

LLMs enable rapid generation of expert judgments for Tier 3 parameter estimation. When experimental data is unavailable, LLM-based elicitation can generate prior distributions that are then updated with any available evidence (Bayesian approach).

As documented in Appendix AN, LLM estimates correlate highly with experimental benchmarks ($r = 0.87$, $p < 0.0001$), validating their use as auxiliary estimators.

9.5 Computational Structural Estimation

Modern computational infrastructure enables estimation of high-dimensional models:

- **Automatic Differentiation**: Gradient computation no longer requires analytical derivation (automatic via TensorFlow, JAX, PyTorch)
- **Bayesian Methods**: Markov Chain Monte Carlo and variational inference enable flexible uncertainty quantification
- **Sparse Matrix Representations**: 144×144 matrices are computationally trivial; 10,000×10,000 matrices are feasible
- **Transfer Learning**: Patterns estimated in one domain transfer to others, reducing data requirements

9.6 The Feasibility Frontier

The empirical program outlined in this paper—simultaneous estimation of context-dependent complementarity matrices across multiple domains—is *now feasible* as of 2024-2026. It remains challenging but is no longer in the realm of science fiction.

The convergence of five technologies enables this:

1. Data availability (billions of behavioral observations)
2. Computational power (high-dimensional estimation routine)
3. Machine learning (pattern detection in residuals)
4. Real-time data (tracking transition dynamics)
5. Large language models (hypothesis generation and auxiliary estimation)

Chapter 10

Falsifiability and Progressive Research Programs

10.1 What Makes a Framework Scientific?

Karl Popper's central criterion is falsifiability: a theory is scientific if there exist conceivable observations that would refute it.

The framework makes testable claims:

1. **Complementarities exist**: $C_{ij} \neq 0$ in typical economic decisions
2. **They are measurable**: Estimation procedures can recover C with acceptable precision
3. **They are context-dependent**: $C = C(\Psi)$ with systematic variation
4. **Reference structures exist**: Coherence $K \approx 1$ implies $C^*(\Psi)$ explains behavior better than naive models

Each claim is falsifiable. If comprehensive testing showed that:

- Preferences are separable ($C_{ij} = 0$ universally), OR
- Complementarities are not measurable with acceptable precision, OR
- Context variables do not explain heterogeneity, OR
- Coherence is far from 1 in most settings

then the framework would be rejected.

10.2 Lakatosian Analysis: Hard Core and Auxiliary Belt

Imre Lakatos proposed that mature scientific programs consist of:

- **Hard Core**: Fundamental principles protected from refutation
- **Protective Belt**: Auxiliary hypotheses that can be modified to accommodate evidence

The EBF framework has:

Hard Core vs. Protective Belt

Hard Core (Protected):

- Preferences have a complementarity structure C
- Behavior depends on context Ψ
- Stability is coherence, not equilibrium

Protective Belt (Adaptive):

- The 8C CORE structure (can be extended to 9C, 10C if evidence warrants)
- The specific context dimensions in Ψ (can be enriched with new dimensions)
- Functional forms of $C^*(\Psi)$ (can be refined with better data)
- Measurement methods (can be improved with new technology)

The hard core is protected by the fact that complementarity is demonstrated across 500+ experiments. The protective belt can be modified—and has been—based on evidence (e.g., adding interaction terms when Type II deviations suggest missing effects).

This makes the framework a progressive research program in Lakatos's sense.

Chapter 11

Conclusion: From Evidence to Prediction and Intervention

11.1 Summary of Findings

This paper synthesized empirical evidence across five research traditions, all pointing to a consistent conclusion:

1. **Complementarities are real.** Five hundred experiments across 8 paradigms demonstrate that preferences are not separable. Utility depends jointly on own and others' payoffs, own consumption and others' consumption, etc. This is robust across replication.

2. **Complementarities are measurable.** Structural estimation methods (Fehr-Schmidt, Bolton-Ockenfels, and modern ML approaches) can recover complementarity parameters with acceptable precision. Tier 1 experimental estimates, Tier 2 meta-analyses, and Tier 3 LLM-assisted estimation provide overlapping paths to parameters.

3. **Complementarities are context-dependent.** Effect sizes vary with stakes, subject pool, country, institutions. This variation is systematic and predictable: $C = C(\Psi)$ where Ψ includes institutional, cultural, and social factors.

4. **Complementarity structures have neural substrates.** Complementarity is not a metaphor but is encoded in brain circuits. Neuroendocrine states, neural activation patterns, and evolutionary biology all support the reality of social preferences.

5. **Structures are nearly universal, parameters vary.** Across 15 societies and diverse institutional contexts, complementarities appear everywhere, but magnitudes vary with

culture and institutions. The structure is universal; the parameters are context-specific.

11.2 Policy Implications

The framework has direct implications for policy intervention:

> *"If preferences contain complementarities, then optimal policy must attend to complementarity structures, not just to mean outcomes."*

Example: Labor market policy. If workers' effort is complementary with perceived fairness of wages (as field evidence suggests), then raising minimum wages may be more effective than economists traditionally predicted. The mechanism is not just income effect but preference complementarity: fair wages trigger reciprocal effort.

Similarly, climate policy that leverages social complementarities (peer effects, norm conformity) may be more effective than price-based instruments alone.

11.3 Research Agenda

The empirical program outlined herein—estimating context-dependent complementarity matrices—is not yet complete. Priority areas for future research:

1. **Cross-Domain Integration**: Estimate complementarities across FEPSDE dimensions simultaneously, not separately
2. **Behavioral Change Dynamics**: Model how complementarity structures adapt during transitions (e.g., post-crisis recovery, institutional reform)
3. **AI and Complementarity**: How do artificial intelligence and automation affect human complementarity structures? Do they enhance or degrade social preferences?
4. **Scalability**: Develop methods for real-time, continuous estimation of context-dependent complementarities in large populations

11.4 Closing Remarks

The Evidence-Based Framework is not merely theoretical architecture. It rests on half a century of empirical evidence—experimental, field, computational, neurobiological—all demonstrating that economic behavior is shaped by preference complementarities and contextual variation.

Technological capabilities emerging 2020-2026 have rendered the empirical program tractable. The framework can be tested, refined, and deployed for prediction and intervention.

The evidence suggests that understanding human economic behavior requires attending to the structure of preferences—not as fixed and separable, but as complementary and context-dependent. This perspective opens new pathways for both theory and practice.

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